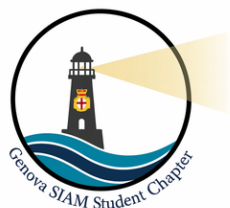




SIAM Chapters Meeting for Young Researchers in Applied Mathematics and Scientific Computing

Politecnico di Milano
January 29-30, 2026



Institutions involved

(in alphabetic order)

Politecnico di Milano, Dipartimento di Eccellenza 2023-2027: polimi.it/eccellenza/



MOX Laboratory, Dipartimento di Matematica, Politecnico di Milano: mox.polimi.it/



INdAM GNCS: www.altamatematica.it/gncs/



Society for Industrial and Applied Mathematics (SIAM): siam.org/



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About the conference

This volume collects the abstracts presented at the SIAM Chapter Meeting in Applied Mathematics. Building on the success of the first Joint GNCS–SIAM Chapters Meeting for Young Researchers in Numerical Analysis and Applied Mathematics, the event aims to foster scientific interaction among early-career researchers in applied mathematics, numerical analysis, scientific computing, and statistical learning.

The scientific program includes 5 keynote lectures by senior researchers, as well as 20 talks by PhD students and postdoctoral researchers. The volume also includes the abstracts of 31 poster contributions, which contribute, through their scientific content, to the overall value of the conference. The abstracts collected herein reflect current research directions and methodological developments in applied mathematics and related fields, highlighting both theoretical and applied perspectives.

The meeting is jointly organised by the Italian SIAM Chapters at Politecnico di Milano (PoliMi), Scuola Internazionale Superiore degli Studi Avanzati (SISSA), Università di Genova (UniGe), Università di Pavia – IMATI (UniPV-IMATI), and Università di Pisa – Scuola Normale Superiore (UniPi-SNS). The event is funded by MUR (italian ministry of university and research), Dipartimento di Eccellenza 2023-2027, GNCS-INdAM and held under the patronage of the Società Italiana di Matematica Applicata e Industriale (SIMAI).

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Keynote talks

Optimisation-driven deep learning for image reconstruction

Luca Calatroni

MaLGa Center, DIBRIS - Università di Genova (Italy)

Recent advances in machine learning and deep learning have revolutionised the study of applied inverse problems and computer vision. Although these data-driven methods often outperform classical reconstruction techniques grounded in tailored modelling and expert knowledge, they frequently lack rigorous mathematical guarantees. In this talk, I will discuss how hybrid approaches that integrate physical models with learned regularisation can bridge this gap, offering a balanced combination of interpretability, reliability, and performance. I will focus in particular on convergent hybrid methods formulated as extensions of classical optimisation schemes (such as proximal gradient methods and bilevel optimisation approaches), in which the physical forward model is enforced explicitly while the prior is learned from data. As an application, I will demonstrate how these techniques can be effectively employed for image reconstruction in fluorescence microscopy, particularly in regimes with limited photon counts.

Low-rank methods in Isogeometric Analysis

Monica Montardini

Università degli studi di Pavia (Italy)

The numerical solution of Partial Differential Equations with standard approaches suffers from the so-called curse of dimensionality: the memory storage and the computational effort grow exponentially with respect to the dimension d of the problem. On a single-patch domain, this is typically $O(n^d)$, where n is the number of degrees of freedom in each univariate direction. As a remedy, in this talk we combine Isogeometric Analysis with low-rank techniques, allowing an overall computational cost and memory footprint that is proportional to n . First, we approximate the equation's coefficients, including those stemming from the geometry mapping, with a sum of few separable functions. We then consider an iterative Krylov solver with tensor rank truncations combined with a novel preconditioning strategy, that is robust with respect to the relevant parameters. The preconditioner is based on the Fast Diagonalization method, that is recast to be compatible to the chosen tensor format. This low-rank technique and the preconditioning strategy can be extended to multipatch geometries, using an overlapping Schwarz method where the subdomains can be defined as unions of neighbouring patches.

Dynamical approximation and sensor placement for the state estimation of conservative dynamics

Cecilia Pagliantini

Dipartimento di Matematica, Università di Pisa (Italy)

This talk focuses on the inverse problem of reconstructing an unknown function u from a finite set of measurements, under the assumption that u is the output of a parametric differential equation with unknown input parameters. Typically, the target function u belongs to an infinite-dimensional Hilbert space and the geometry of the solution set M is not known a priori. One way to reduce the complexity of the problem is to approximate M by a linear, finite dimensional subspace V using model order reduction, and to search for an approximation $u^* \in V$ to the state u . It is however known that such an approach becomes ineffective for approximating the dynamics of conservative systems. In this talk we will discuss how to address inverse problems for transport and wave phenomena in the framework of Hamiltonian systems. The method combines symplectic dynamical low-rank approximation to update V with dynamical placement of the sensors to ensure an accurate reconstruction at all times.

Challenges and advances in multiphysics coupling, with applications in cardiac modeling

Francesco Regazzoni

MOX, Dipartimento di Matematica - Politecnico di Milano (Italy)

Multiphysics coupling arises in many scientific and engineering applications and poses significant challenges in balancing computational efficiency with accuracy. Examples include sub-physics characterized by different requirements in terms of mesh size and polynomial degree, which call for transferring coupling variables across independent meshes in an effective and structure-preserving manner; time-integration instabilities that emerge when loosely coupled schemes are employed as a more convenient alternative to fully monolithic approaches; and spatial accuracy issues that arise when dealing with non-matching meshes. In this talk, we consider cardiac simulation as a paradigmatic example of a complex multiphysics problem, both because of its societal impact and because of the numerous numerical challenges it raises. Indeed, clinical translation in this context is often hindered by the computational costs, which prevent a sustainable integration of in silico simulations within healthcare systems.

How to Preserve Moving Equilibria for Hyperbolic Balance Laws with the Global Flux Quadrature

Davide Torlo

Università di Roma La Sapienza (Italy)

Many conservation and balance laws admit families of moving steady states that are crucial to preserve at the numerical level. While several techniques have been developed to achieve this in one dimension, few are available for multidimensional problems, and those often apply only to linear cases. The so-called Global Flux approach integrates multiple terms into a single physical flux, yielding a unified differential operator acting on a more complex flux function. This formulation enables highly accurate discrete equilibria that correspond to exact continuous steady states. In finite element methods, these discrete equilibria exhibit superconvergence properties, allowing for very precise perturbation analyses. Despite its name, the Global Flux method leads to compact numerical stencils and can be reformulated as specific discrete derivative and source terms. We apply the method to various systems — including the acoustic, shallow water, Euler, and Maxwell equations, with and without source terms — demonstrating its ability to preserve a broad class of genuinely multidimensional equilibria.

Participant contributions

CAuSe–AI: Evaluating the Reasoning of Deep Learning Models

Barbara Tarantino

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In scientific and clinical domains, a model is trustworthy only if its internal computations remain coherent with the mechanistic structure that governs the system it reconstructs. When this structure can be formalised as a directed acyclic graph (DAG), assessing such coherence becomes essential. Existing methodological tools fall short: explainability methods quantify associations, causal-learning approaches infer graphs, but neither determines whether a trained model actually computes in accordance with the prescribed causal mechanisms. We address this gap with CAuSe–AI, a framework that combines three key components: (i) a hierarchy of graph-informed architectures; (ii) perturbations designed to preserve empirical dependencies while assessing functional influence; (iii) an operator Φ^* that subtracts the contribution of unconstrained approximation, isolating the mechanistic component learned by the model. From Φ^* , four normalised metrics (fidelity, stability, robustness, and fairness) capture complementary aspects of structural behaviour: how influence concentrates on true edges, how reproducible these allocations are, how resilient they remain under noise, and how structural attention evolves as graph complexity increases. Applied to the Sachs signalling network and to synthetic DAGs of increasing dimensionality, CAuSe–AI reveals reproducible mechanistic regimes: sparse systems support sharply focused pathways, whereas denser or hub-dominated topologies attenuate fidelity but preserve distinctive stability–robustness signatures. These patterns show how causal constraints can shape internal computation while maintaining high reconstruction accuracy. Beyond descriptive insights, CAuSe–AI enables mechanistic auditing, shortcut detection, verification of domain coherence, and principled comparison of architectures. By shifting evaluation from deep-learning outputs to the organisation of the computations that generate them, it provides a solid foundation for deploying AI systems that are both accurate and mechanistically credible in engineered and scientific settings.

Optimal Convergence in Least-Squares Isogeometric Collocation

Maria Roberta Belardo

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Isogeometric collocation (IgA-C) enforces PDEs in strong form at selected sites of smooth spline spaces, avoiding element integration and reducing assembly cost relative to Galerkin IgA. Its accuracy is nevertheless highly sensitive to the collocation nodes: standard choices such as Greville abscissae exhibit suboptimal convergence compared to the Galerkin reference. Several contributions have explored collocation point selection through its relationship with Galerkin discretizations and superconvergence. Variational collocation introduces Cauchy–Galerkin points, defined as zeros of the Galerkin residual,

and proposes element-invariant approximations for their computation. Related results show that collocation at Galerkin superconvergent locations can recover optimal L2 convergence for odd spline degrees in square collocation schemes. Building on this context, recent results show that a simple structural feature, i.e. local symmetry of the interior sites within each knot span, is correlated with optimal-order behavior in global least-squares IgA-C. For spline degrees $p=3-8$, ensembles of elementwise symmetric perturbations of interior collocation points are tested in overdetermined least-squares solves of standard second-order elliptic benchmarks. For $p4$, symmetric configurations systematically achieve Galerkin-like convergence, with L2 errors approaching order $p+1$ and H1 errors approaching order p . Guided by these observations, an element-invariant symmetric selection of interior collocation points is proposed for even degrees, mitigating the odd-even discrepancy observed with standard choices. Validation in 1D and in 2D second order problems on mapped geometries confirms robustness across refinements and boundary treatments.

Calibration of a Mathematical Model for Cardiac Perfusion

Viola Cusumano

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This study considers a mathematical model for cardiac perfusion able to quantify myocardial blood flow (MBF), a quantity of utmost importance in the diagnosis of coronary artery disease. We used a coupled model integrating Navier-Stokes equations for large coronary flow with a three-compartment Darcy model for intramyocardial hemodynamics, incorporating vessel distensibility. The most challenging issue related to this model relies on the calibration of its parameters (Darcy permeabilities and conductivities), which are highly dependent on the patient and on their clinical conditions. For the calibration, patients were stratified into three stenosis-based groups and suitable values of the parameters were extracted from MBF maps obtained from stress-CTP of 9 “calibrating” patients (3 per groups). Specifically, model parameters are adjusted according to the stenosis degree and the volume of the perfusion region, by means of suitable empirical laws. We validated the proposed calibration on 100 patients, covering all the 3 groups, performing numerical simulations with personalized aortic inlet pressure in geometries reconstructed from cCTA scans. Measured (M) and simulated (S) MBF maps were compared in each of the 17 segments into which we divided the left ventricle, computing the percentage of myocardial volume with MBF below clinical thresholds. The results showed a very promising agreement for all the groups between our blind predictions and the clinical stress-CTP data, with sensitivity variable greater than 0.7, i.e with a very good ability to identify pathological regions. All the numerical experiments were conducted with the Finite Elements library LifeX developed at Politecnico di Milano.

An integrated 1D-0D model of the cardiorespiratory system with local autoregulation

Caterina Dalmaso

Università di Trento (Italy)

We present a model of the cardiorespiratory system that includes a 1D-0D description of the cardiovascular system, a 0D description of lung mechanics, a 1D-0D description of gas transport and exchange in the systemic and pulmonary vessels, and local autoregulatory mechanisms. Simulations are performed on two 1D arterial-venous network models characterized by different degrees of anatomical

detail, coupled to a 0D description of lung mechanics. These networks are calibrated to ensure comparable cardiac output, mean arterial pressure, and flow rates in selected arteries. Respiratory gases are modeled as passive scalars transported by blood, enabling investigation of how vascular network resolution influences tracer dynamics. Specifically, we analyze the time required for a bolus injected into the antecubital vein to reach the aortic arch and cerebral vessels. Gas exchange is simulated both at the level of the lungs, assuming instantaneous diffusion across the hemato-alveolar membrane, and between peripheral capillaries and tissues, under the assumption of a homogeneous blood-tissue compartment. The gas exchange module is calibrated accounting for the anatomical detail provided by the arterial network: we consider five gas consumption/production rates, defined for a baseline state based on Fick’s principle, and twenty-five perfused tissue regions, each characterized by a specific volume. Additionally, we include a description of basic autoregulation phenomena, which capture metabolic responses to blood gas concentrations and myogenic responses to flow. Several examples are presented to demonstrate the predictive capabilities of the model and to explore physiologically relevant scenarios.

Deep Equilibrium models for Poisson Inverse Problems via Mirror Descent

Christian Daniele

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Inverse problems in imaging arise in a wide range of scientific and engineering applications, including medical imaging, astrophysics, and microscopy. These problems are inherently ill-posed, requiring advanced regularization techniques and optimization strategies to achieve stable and accurate reconstructions. In recent years, hybrid approaches that combine deep learning and variational methods have gained increasing attention. Well-established techniques include Algorithmic Unrolling, Plug-and-Play methods, and Deep Equilibrium Models. The latter are networks with fixed points, which are trained to match data samples from a training dataset. In this work, we focus on Deep Equilibrium Models to learn a data-driven regularization function for Poisson inverse problems, using the Kullback-Leibler divergence as the data fidelity term. To effectively handle this fidelity term, we employ Mirror Descent as the underlying optimization algorithm. We discuss theoretical guarantees of convergence, even in non-convex settings, incorporating a backtracking strategy, along with key aspects of training this class of models. To validate our approach, we evaluate its performance on a deblurring task with different kernels and varying levels of Poisson noise.

Region-Constrained RBPF for MEG source localisation

Valeria Fiori

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In the field of neuroscience, one of the most interesting problems is identifying neural activity. Among the various methods used to obtain useful data for this purpose, magnetoencephalography (MEG) is a non-invasive technique capable of measuring the magnetic field produced by brain activity on the scalp, ensuring high temporal precision. To solve the inverse problem of localising neural sources, various methods have been proposed, including those based on the Bayesian approach. In these approaches, each variable is considered random and, unlike classical regularisation methods, a single point estimate of the unknown is not obtained, but rather a probability distribution (posterior) that also provides

an estimate of the uncertainty of the solution. In particular, the algorithm described is the Rao-Blackwellised Particle Filter (RBPF), which is particularly advantageous for the analysis of dynamic models and, compared to the classic Particle Filter, provides estimates with lower variance thanks to the marginalisation of certain variables. However, one limitation of RBPF is the possibility that, over time, large shifts in neural sources may be observed, which are clinically implausible. To solve this problem, brain atlases that segment the brain into various areas were used. It was therefore imposed that a neural source cannot leave its region of origin. The proposed algorithm was tested on synthetic data using different brain atlases. The results obtained show a significant improvement in the accuracy of brain activity reconstruction compared to unconstrained methods.

Dynamical Reduced Order Approximation of Wasserstein Gradient Flows

Isabella Carla Gonnella

Scuola Internazionale di Studi Superiori Avanzati, Trieste (Italy)

This work introduces a novel dynamical reduced-order approximation method for partial differential equations (PDEs) that exploits the geometric structure of the solution manifold to construct an adaptive reduced representation. The proposed approach evolves the parametrization of the reduced basis through suitably constructed systems of ordinary differential equations, enabling the approximation space to adapt dynamically. The evolution of the approximation space is shown to be optimal with respect to the metric induced in the tangent space of the solution manifold, thereby ensuring an efficient and accurate representation of the underlying dynamics. The method is illustrated in the context of Wasserstein gradient flows, which arise in a wide range of applications spanning advection-diffusion-reaction PDEs and Optimization, and whose numerical treatment becomes particularly challenging in high-dimensional settings. In this regime, the numerical approximation of Wasserstein gradient flows is challenged by the curse of dimensionality, the prohibitive cost of optimal transport computations, and sampling inefficiencies in the representation of evolving probability measures. These difficulties are compounded by nonlinear interactions and the simultaneous presence of diffusive and transport-dominated dynamics. In contrast, the proposed dynamical approach does not require a mesh, attains high accuracy with a small number of basis functions, and requires only a single sampling of the initial condition. The proposed framework contributes to the development of a general nonlinear reduced-order model (ROM) capable of accurately describing a wide range of physical scenarios. Its effectiveness is demonstrated on well-known Wasserstein gradient flows, including linear and double-well Fokker-Planck equations and the Porous Medium equation.

On the Role of Overparameterization in Optimization Dynamics

Hippolyte Labarrière

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Gradient-based methods successfully train highly overparameterized models, despite the nonconvex nature of the underlying optimization problems. In this work, we study the dynamics of gradient flow in such regimes to better understand this phenomenon. We examine settings where overparameterization induces a geometry favorable to optimization, effectively allowing gradient-based algorithms to behave as if the problem were convex. We discuss the structural conditions and initialization scales that enable

this behavior, providing theoretical guarantees for convergence to global minima. Our analysis offers a perspective on how overparameterized landscapes can facilitate efficient training.

A p-adaptive high-order polytopal method for modelling neuronal electrophysiology

Caterina Beatrice Leimer Saglio

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Traveling wave-like phenomena arise in a wide range of biological processes, including electrical signal propagation in the nervous system and brain tissue. Accurately simulating these phenomena poses significant numerical challenges, as sharp and rapidly moving wavefronts require high spatial and temporal resolution to be properly captured. Such requirements often lead to prohibitive computational costs, especially in large and heterogeneous domains. The evolution of the transmembrane potential is governed by steep, fast wavefronts propagating across multiple brain regions, often following preferential axonal pathways. These dynamics emerge from complex multiscale interactions, where rapid ionic exchanges at the cellular level generate electrical signals that propagate through anisotropic and heterogeneous tissues. High-order discontinuous Galerkin methods on polygonal and polyhedral meshes (PolyDG) offer great flexibility and accuracy for such problems, but their computational cost can remain significant when uniform high-order discretizations are used. To address this limitation, we propose a p-adaptive PolyDG strategy that exploits the intrinsic traveling wave structure of the solution. In particular, the transmembrane potential exhibits strong spatial variations localized near the propagating wavefront, while the solution remains nearly stationary elsewhere. Our approach consists in the design of efficient a posteriori error indicators capable of accurately detecting the wavefront location. These indicators drive fully automatic local adjustment of the polynomial degree, allowing higher-order approximations only where they are truly needed. We present numerical results that validate the effectiveness of the proposed method and demonstrate its application to the simulation of epileptic events in heterogeneous brain domains, including grey and white matter. The results show that polynomial adaptivity significantly reduces the total number of degrees of freedom and the overall computational cost while maintaining high-order accuracy throughout the simulation.

An entropic lattice boltzmann method for wettability control via coupled boundary-force interpolation

Marco Desiderio Loffredo Senesi

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In a multiphase fluid system, the balance between cohesive and adhesive non-ideal forces determines whether the denser phase spreads over or is repelled from a boundary surface. Wetting can either promote or hinder mass transfer, affecting both natural phenomena, such as insect survival in harsh habitats, and the performance of engineering devices, such as in advanced energy-conversion systems. Predictive simulations require a physically consistent numerical framework that remains robust on coarse grids. The lattice Boltzmann method (LBM) is a suitable candidate, solving the semi-incompressible Navier-Stokes equations via the mesoscopic dynamics of particle colliding and streaming on a lattice. In the Shan-Chen pseudopotential model, non-ideal fluid-fluid and fluid-solid interactions, being cohesion and adhesion, are introduced via body forces depending on an effective mass: this results in

a spontaneous liquid-gas phase separation below a critical temperature and in the emergence of surface tension effects. However, existing multiphase-wetting schemes typically suffer from low accuracy and instability at large density ratios between phases. They also treat cohesive and adhesive forces separately and suffer from mass leakage on unfitted boundaries. To tackle these problems, a unified multiphase LBM for multiphase flows and wetting is proposed. The scheme combines (i) an advanced central-moments-based collision operator with entropic stabilization; (ii) a unified pseudopotential force that allows treatments developed to refine cohesive interactions to be consistently extended to adhesive interactions; and (iii) an improved mass-refill strategy for the interpolated bounce-back scheme, the Unidirectional Coupled Boundary Wettability (U-CBW) approach, which exploits local interaction-force data to enforce strict mass conservation. Furthermore, a Bidirectional CBW (B-CBW) scheme is introduced, capable of imposing wall-location sensitivity within a compact force stencil. The method is assessed on multiphase wetting benchmarks, including static and dynamic droplet tests and wetting on flat and curved substrates. The accuracy, stability, and mass conservation properties at the boundaries are discussed.

Poroelectricity for intrinsically incompressible constituents

Chiara Lonati

Politecnico di Torino (Italy)

In standard approaches to the behaviour of a poroelastic solid with fluid flowing through it, one relies on Biot's theory and on the asymptotic homogenization method; however, in its usual applications, the elastic material is typically assumed to be compressible. We treat the derivation of the equations of Biot's poroelasticity from the microstructure in the limit case of incompressibility of both an isotropic, linear elastic solid and a low Reynolds' number Newtonian fluid flowing through its pores. This fluid-structure interaction problem requires the introduction of a pressure for the solid and the fluid phase; a proper formulation cannot be obtained just by letting the Poisson ratio of the two constituents tend to 0.5. Thus, we impose a divergence-free constraint in both materials and we employ the asymptotic homogenization technique to derive a macroscale system of PDEs, with coefficients which are to be computed by solving non-standard periodic cell problems at the pore-scale. Moreover, we retrieve the equivalence between the change in volume of the porous solid and the volume of fluid exchanged.

TMATDG: applying TDG methods to multiple scattering via T-matrix approximation

Armando Maria Monforte

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We present TMATDG, a MATLAB package for the numerical solution of two-dimensional multiple scattering problems of time-harmonic acoustic waves, governed by the Helmholtz equation. The approach is based on the T-matrix framework, which reduces a multiple scattering problem to the solution of independent single scattering problems, whose results are then coupled through linear relations. A key feature of the approach is that the T-matrix, which represents the scattering properties of a single obstacle, depends only on the obstacle shape and material parameters and can therefore be computed

once, stored, and reused, yielding substantial computational savings for ensembles containing repeated shapes.

Preconditioning techniques for fictitious domain formulations of elliptic interface problems

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We present a novel augmented Lagrangian (AL) preconditioner for solving linear systems arising from finite element discretizations of elliptic interface problems, where the coefficients of the governing partial differential equation may jump across the interface between different subdomains. Such problems are central in many scientific and engineering fields because of their wide-ranging applications, such as bio-science and fluid–structure interaction, where the fluid and solid domains exhibit distinct physical properties across the interface. The approach builds on the Fictitious Domain with Distributed Lagrange Multipliers (FD-DLM) formulation, which, while effective for complex geometries, leads to large and ill-conditioned linear systems. For this reason, we design preconditioners that are robust with respect to both mesh refinement and possibly large coefficient jumps. We achieve this by extending to the FD-DLM framework the AL strategy that we recently proposed for boundary-supported Lagrange multipliers. Additionally, we introduce a computationally cheaper block-triangular variant that offers easier implementation while maintaining low iteration counts. Numerical experiments using the C++ finite element library deal.II demonstrate the robust performance of this variant with respect to both mesh refinement and large coefficient jumps.

Fast physics-informed spatio-temporal smoothing with time-indexed elliptic operators

Alessandro Palummo

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We study the problem of modeling spatial data that also evolve along a non-spatial dimension, such as time or frequency. When data are supported on non-trivial spatial domains, standard spline-based smoothers become unsatisfactory. Moreover, spatio-temporal phenomena typically display anisotropic and non-stationary behavior governed by underlying dynamical laws. Physics-informed statistical learning provides a unified framework that combines observed data with prior physical knowledge modeled through a Partial Differential Equation. Although one could, in principle, describe the joint space–time dynamics through a single space–time differential operator, such a formulation may be unjustified when knowledge of the full space–time dynamics is incomplete. In this work, we therefore address the smoothing of space–time data when complete knowledge of the joint dynamics is not available, modeling this partial knowledge through a time-indexed family of elliptic operators. Using the viewpoint of optimal control theory, we prove the well-posedness of the estimation problem when the PDE coefficients vary in space and time. Moreover, to address the computational challenges posed by high-resolution space–time data, we depart from existing literature based on space–time Galerkin formulations and instead propose a parallel Waveform Relaxation procedure for the numerical solution of the estimation problem. We demonstrate the effectiveness of the proposed methodology on massive data through a real application.

A Preconditioned Version of a Nested Primal-Dual Algorithm for Image Deblurring

Giuseppe Scarlato

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Regularization techniques for image deblurring problems typically consist of a smooth term and a potentially non-smooth convex term. A common approach to solving these problems is using proximal gradient methods. To accelerate the convergence of these first-order iterative algorithms, strategies such as variable metric methods have been introduced in the literature. In particular, regarding the variable metric strategy proposed for the Nested Primal–Dual Iterated Tikhonov method (NPDIT), we proved that, for linear inverse problems, this approach can be reinterpreted as a right preconditioning method. Consequently, we explore an inexact left-preconditioned version of the same proximal gradient method. We prove the convergence of the new iteration to the minimum of a model where the norm of the data fidelity term depends on the preconditioner. The numerical results show that left and right preconditioning are comparable in terms of the number of iterations required to reach a prescribed tolerance, but left preconditioning needs much less CPU time, as it involves fewer evaluations of the preconditioner matrix compared to right preconditioning. The quality of the computed solutions with left and right preconditioning are comparable. Finally, we propose some non-stationary sequences of preconditioners that allow for fast and stable convergence to the solution of the variational problem with the classical euclidean norm on the fidelity term.

A p-adaptive high-order polytopal method for modelling neuronal electrophysiology

Giacomo Speroni

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Soilless cultivation is an emerging trend in modern agriculture that substitutes common soil with engineered growing substrates whose mechanical and transport characteristics ensure proper root support, aeration, and nutrient delivery. However, conventional organic and inorganic growing media offer limited controllability and often raise sustainability concerns, motivating the development of engineered alternatives. To address the highlighted limitations, inspired by the efficiency of natural porous microstructures (e.g., sponges, bones, wood), we develop a computational-to-manufacturing framework to design optimized 3D-printed porous scaffolds for soilless cultivation. The proposed architectures are conceptualized as lattice structures, whose effective macroscopic properties are governed by the geometry of the underlying periodic microstructure. Accordingly, the design phase is formulated as a topology optimization problem aimed at systematically tailoring the microstructural geometry under coupled mechanical and fluiddynamic requirements and design constraints. To this aim, we extend the microSIMPATY algorithm, proposed in in a 2D context, to a fully 3D multi-physics setting, coupling densitybased SIMP topology optimization with anisotropic mesh adaptation. The proposed topology optimization algorithm is first validated through a set of numerical test cases aimed at assessing the behavior and robustness in a multi-physics setting. Leveraging the obtained results, selected optimized topologies are converted into explicit lattice geometries as STL-ready digital models. These models

enable both in silico root growth simulations, performed by extending the RootBox modeling environment, and the fabrication of 3D-printed physical substrates. The integrated approach allows the same digitally engineered porous architectures to be used consistently for numerical validation and prototype realization, providing a unified pathway from multi-physics optimization to simulation-driven assessment and 3D-printed devices for soilless cultivation.

Latent Dynamics Graph Convolutional Networks for model order reduction of parameterized time-dependent PDEs

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Graph Neural Networks (GNNs) have emerged as powerful tools for nonlinear Model Order Reduction (MOR) of time-dependent parameterized Partial Differential Equations (PDEs). However, existing methodologies struggle to combine geometric inductive biases with interpretable latent dynamics, overlooking dynamics-driven features or disregarding geometric information, respectively. In this work, we address this gap by introducing Latent Dynamics Graph Convolutional Network (LD-GCN), a purely data-driven, encoder-free architecture that learns a global, low-dimensional representation of dynamical systems conditioned on external inputs and/or parameters. The temporal evolution is modeled in the latent space and advanced through time-stepping, allowing for time-extrapolation, and the resulting trajectories are consistently decoded onto geometrically parametrized domains using a GNN. Our framework enhances interpretability by enabling the analysis of latent trajectories and supports zero-shot prediction through interpolation in the latent space. The methodology is mathematically validated via a universal approximation theorem for encoder-free architectures, and numerically tested on complex computational mechanics problems involving physical and geometrical parameters, including the detection of bifurcating phenomena for Navier-Stokes equations.

Sparse sensing is all you need: real-time simulation and optimal control of high-dimensional parametric systems

Matteo Tomasetto

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Reduced order modeling (ROM) is of paramount importance for efficiently monitoring and controlling complex systems in multi-scenario contexts, enabling early risk detection and informed data-efficient decision-making. However, conventional ROM strategies are typically limited to known and constant parameters, inefficient for nonlinear and chaotic dynamics, and blind to the actual system behavior. In this talk, we present a sensor-driven Reduced Order Modeling strategy based on SHallow REcurrent Decoder networks (SHRED-ROM). Specifically, motivated by the well-known method of separation of variables, we consider the composition of a sequence model, which encodes the temporal dynamics of limited sensor measurements in multiple scenarios, and a decoder model, that reconstructs the corresponding high-dimensional spatio-temporal field. To enhance computational efficiency and memory usage, the snapshots dimensionality is reduced by data- or physics-driven basis expansions, allowing for compressive training of the networks with minimal hyperparameter tuning. By employing SHRED-ROM to control problems, it is possible to efficiently retrieve distributed control actions starting from sparse state readings in multiple scenarios. Through applications on nonlinear and chaotic dynamics, we show that SHRED-ROM (i) accurately reconstructs and controls challenging dynamics in multiple

scenarios with minimal sensor requirement, independently on sensor placement, (ii) can cope with both physical, geometrical and time-dependent parametric dependencies, while being agnostic to their actual values, (iii) can accurately estimate unknown parameters, and (iv) can deal with different data sources, such as high-fidelity simulations, coupled fields and videos.

Stochastic algorithm for the 3D Computed Tomography (CT) problems

Ilaria Trombini

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A discrete linear inverse problem can be expressed as $y = Kx + \varepsilon$, where $x \in \mathbb{R}^n$ is the unknown to be reconstructed, $K \in \mathbb{R}^{m \times n}$ is the discretized forward operator, $y \in \mathbb{R}^m$ the measured data, and ε Gaussian noise. In Computed Tomography (CT), K corresponds to the Radon transform. Its discrete formulation is strongly ill-conditioned, making direct inversion unreliable. A common approach to stabilizing the reconstruction is variational regularization, in which one solves

$$\min_{x \in \mathbb{R}^d} D(Kx, y) + \beta R(x),$$

where D measures the discrepancy between Kx and the data, R is a regularization functional preserving essential image features, and β balances fidelity and regularity. First-order deterministic optimization schemes have long been the standard tools to address such problems thanks to their simplicity and proven convergence. However, the increasing size and complexity of modern CT datasets make their computational cost prohibitive. For this reason, the focus of this work shifts toward stochastic optimization. Stochastic algorithms—widely used in machine learning—significantly reduce the cost per iteration by operating on subsets of the data, while still ensuring convergence under suitable assumptions. This talk presents a strategy for applying stochastic optimization methods to 3D CT reconstruction with an adaptive management of key hyperparameters. Algorithms originally developed for machine learning and deep learning are adapted to the imaging context by analyzing mini-batch selection, step-size adjustment, and the preservation of theoretical convergence guarantees. Numerical experiments will demonstrate the performance and advantages of the proposed approach.

PINNs and ROMs for predictive maintenance of cultural heritage

Carmine Valentino

Università di Salerno (Italy)

Preserving cultural heritage often demands predictive tools capable of representing intricate physical mechanisms and their evolution over time. In this work, we introduce a digital framework based on Scientific Machine Learning to enhance the analysis and maintenance planning of cultural assets. The proposed solution is designed to efficiently solve parameter-dependent partial differential equations by combining Physics-Informed Neural Networks and Reduced Order Models. PINNs enforce governing equations and boundary/initial constraints directly during the learning process, while ROM methodologies deliver fast surrogate representations for high-dimensional systems. We also discuss the practical implications of blending data-driven and physics-based modules, focusing on issues such as sampling design, generalization capability, and confidence in model outputs when dealing with

multi-scale phenomena. To evaluate the approach, we consider simulated case studies built on time-dependent, parameterized PDEs. The results highlight the potential of hybrid PINN-ROM strategies to provide computationally light yet physics-consistent approximations, thereby enabling reliable decision-support tools for cultural heritage conservation and predictive maintenance. This is a joint work with F. Colace (UNISA), D. Conte (UNISA), F. Pichi (SISSA), and G. Rozza (SISSA).

Poster presentations

Regularization approaches in Image Scanning Microscopy (ISM)

Sofia Agostoni

Università degli studi di Genova (Italy)

Image Scanning Microscopy (ISM) is a modern technique that enhances resolution by combining multiple measurements, particularly under low-photon conditions. We address the reconstruction task by formulating it as a Poisson inverse problem. A maximum a posteriori framework is employed, combining a multi-detector KL-divergence data fidelity term with explicit regularization, including smoothed total variation and Tikhonov/Sobolev. Reconstruction is performed using projected gradient descent, which makes the approach amenable to deep-learning-based variants enforcing regularisation through the use of a pre-trained image denoiser. The regularization strength is selected via the Nearly Exact Discrepancy Principle for Poisson noise, adapted to the ISM framework and the low photon count, relying solely on measurement data without requiring ground truth, and yielding stable, high-quality reconstructions in challenging regimes.

Infinite-Dimensional Generative Compressed Sensing via Coherence-Based Sampling

Paolo Angella

Università degli studi di Genova (Italy)

We develop a theoretical framework for generative compressed sensing in infinite-dimensional Hilbert spaces. Signals are modeled as outputs of generalized generative networks mapping finite-dimensional latent variables into L^2 , and are recovered from noisy, randomly subsampled measurements of a unitary operator. We introduce an infinite-dimensional notion of local coherence that quantifies the interaction between the generator and the measurement basis, leading to optimal coherence-based sampling strategies. We prove a generalized Restricted Isometry Property with sample complexity independent of ambient resolution. Numerical experiments on PDE-based inverse problems confirm the theory and demonstrate improved recovery over uniform sampling, particularly in undersampled regimes.

Barzilai-Borwein Step Sizes for Incremental Gradient Methods in Non-Smooth Optimization

Narges Araboljadidi

Università degli studi della Campania (Italy)

Incremental gradient methods have become central to large-scale optimization, particularly when the objective function can be decomposed into a sum of component terms, as is commonly encountered in distributed computing and machine learning. Although Barzilai-Borwein (BB) step size rules are known to enhance convergence in smooth optimization, extending their benefits to non-smooth composite problems with regularization remains non-trivial. This work investigates the incorporation of BB-type step size strategies into incremental proximal gradient algorithms for sparse optimization. We consider both classical BB rules and recent variants tailored to the presence of non-smooth terms, and evaluate their practical behavior when component gradients are processed sequentially in distributed settings. Our analysis highlights how the incremental structure interacts with adaptive step selection, shedding light on conditions under which BB-based strategies can improve efficiency and robustness in composite optimization.

Fine-tuning a deep learning-based system for predicting drivers' lane change behavior

Maryam Bavafaye Semiromi

Università degli studi di Genova (Italy)

Anticipating a driver's lane change several seconds before lateral motion begins is a key enabler for advanced driver assistance systems (ADAS) that warn early without overwhelming users with false alerts. We develop and evaluate a modular deep-learning pipeline for early lane-change anticipation that jointly performs binary intent recognition for change vs keeping the lane, and time to lane change estimation, while offering feature attributions for transparency (SHAP analysis), and support for online personalization. Data are collected in a CARLA-based highway testbed with 60 km of two lane roads, straight and curved segments. Participants with diverse demographics complete baseline (lane keeping) and overtaking drives, yielding multivariate time series of ego dynamics, lane geometry, indicators, and relative kinematics of surrounding vehicles sampled at 10Hz, windowed for sequence modelling. We study two alternative backbones. The first cascades a CNN-BiLSTM intent classifier with a BiGRU time to lane change regressor. The second uses a single BiLSTM regressor mapping to a continuous target that jointly encodes timing and lane keeping, from which binary intent is induced. To interpret the models, we use a shared SHAP-based feature attribution pipeline that identifies which input variables most influence the predictions, both for individual samples and on average, and confirms that the systems rely on driving plausible cues including headway to the adjacent lane lead vehicle and indicator state dominate intent, while steering angle, lateral acceleration, and distances to lane markings dominate timing. Both systems achieve intent accuracies around 0.96 with lane-change timing RMSE = 0.19. An online fine-tuning strategy, triggered after completed maneuvers, adaptively retrains on recent lane changes and stable driving, improving accuracy and robustness to individual

driving styles. Although the study is simulation-based, we discuss extensions toward Liquid Neural Networks to endow the encoder with input-adaptive temporal dynamics under distribution shift.

Personalized Multiscale Cardiovascular Modelling of Electrical Dyssynchrony Using Constraint Bayesian Optimization

Ester Bergantin

Università di Trento (Italy)

Cardiac electrical asynchrony, caused by conduction disorders such as left bundle-branch block (LBBB), leads to heterogeneous myocardial contraction and can impair cardiac pump function. In this context, cardiovascular modelling is a powerful tool for investigating dyssynchronous heart failure and its treatments. The goal of our research is to develop a personalized cardiovascular modelling pipeline to study the impact of electro-mechanical dyssynchrony on cardiac function. We compute patient-specific activation maps in LBBB subjects using a three-dimensional biventricular electrophysiology model. These activation maps are then integrated into CircAdapt, a lumped mechanical and haemodynamic model, to predict the consequences of dyssynchrony on regional myocardial mechanics and global pump performance. A key aspect of our pipeline is the probabilistic learning strategy adopted to separately calibrate the two coupled models using non-invasive clinical data. We employ a constraint Bayesian optimization framework, which iteratively builds surrogate probabilistic models of both the objective function and the feasibility constraints, and selects new evaluations by balancing exploration and exploitation.

Learning Vascular Hemodynamics: Physics- Informed GNNs Inference from Capillary Networks to Pulsatile Flow

Paolo Botta

MOX, Dipartimento di Matematica - Politecnico di Milano (Italy)

Simulating microvascular blood flow in realistic vascular networks is computationally challenging due to geometric complexity, multiscale structure, and nonlinear blood rheology. We propose a graph-based, physics-informed learning framework that employs Graph Neural Networks (GNNs) as surrogate models for microcirculatory hemodynamics. The approach accurately predicts pressure, velocity, and hematocrit in capillary networks under steady-state conditions while ensuring physical consistency through appropriate loss functions. Building on this framework, we introduce a complementary GNN architecture for time-dependent hemodynamics that can learn and propagate pulsatile flow dynamics in arteriolar networks. The proposed models are generalizable to complex vascular topologies and enable fast and scalable simulations for biomedical applications.

Neural Markov chain Monte Carlo: Likelihood density estimation via normalizing flows and variational autoencoders

Giacomo Bottacini

MOX, Dipartimento di Matematica - Politecnico di Milano (Italy)

We present a new methodology for solving inverse problems in a variational framework that not only yields predictive results but also provides a quantification of uncertainty. To achieve this, we developed a specific MCMC framework to estimate the posterior distribution of the system parameters that generated the given observation, employing a Differential Evolution Metropolis sampling method. In complex problems, where the likelihood function is typically intractable or unavailable, we introduced a Normalizing Flow structure, specifically using Real-valued Non-Volume Preserving transformations (RealNVP). The scaling and translation functions of the affine coupling layers are modeled by neural networks that are conditioned on the label, allowing the model to capture complex posterior distributions. The inference process begins with the observation being processed by a Variational Autoencoder (VAE). This step reduces the dimensionality of the input to be handled by the RealNVP and extracts the most relevant features for parameter estimation. To further enhance the informativeness of the latent space, we added dense layers that sampled from the latent distribution and aimed to predict the correct label. By including this supervised loss term alongside the standard VAE losses, the latent space becomes more structured and informative for the downstream inference task. The proposed methodology is validated on two case studies: a steady-state groundwater flow problem governed by Darcy’s law and a railway bridge structure. In the first case, the method shows strong performance in reconstructing the spatial distribution of the conductivity field. In the second case, it effectively estimates both the location and severity of structural damage.

When Deep Autoencoders meet the Eckart-Young-Schmidt theorem: error analysis and initialization strategies

Simone Brivio

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Deep symmetric autoencoders are a recently proposed framework which bridge the expressive power of nonlinear autoencoders with the solid mathematical structure of linear reduction methods. In this work, we put forth an original framework to provide error estimates for deep symmetric autoencoders, which ultimately leads us to the design of a novel initialization strategy inspired by the Eckart-Young-Schmidt Theorem. We complement our theoretical analysis with numerical tests comprising classical model reduction benchmarks and applications.

Numerical frugality in optimization: mixed-precision Newton’s method

Giuseppe Carrino
ENS Lyon (France)

In Machine Learning optimization first-order methods dominate in practice, while higher-order approaches such as Newton’s method offer higher accuracy and faster convergence, but are often avoided due to their high computational cost. In our work, we analyze the finite-precision errors propagation in Newton’s step, proposing a convergence theorem for mixed-precision Newton optimizers, including “Quasi” and “Inexact” variants. The theorem not only guarantees convergence for not too ill-conditioned problems, but also allows predicting in advance the accuracy achievable by the optimizers. Numerical experiments show that our method sacrifices low, or none, accuracy, while being less computationally expensive than classical Newton.

PINA: A Unified Framework for Scientific Machine Learning

Dario Coscia, Giovanni Canali
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Scientific Machine Learning (SciML) is reshaping how complex scientific systems are modelled and simulated. As problems grow more complex, flexible and scalable tools are increasingly essential. PINA is an open-source library designed to make SciML development fast, intuitive, and scalable. Built on PyTorch and Lightning, and fully compatible with PyTorch Geometric, PINA provides a unified framework for tackling diverse scientific challenges, from learning solutions to PDEs and constructing machine learning force fields, to simulating dynamics and modelling deformations. By combining the power of PyTorch with SciML-focused tools, PINA empowers researchers to experiment and deploy advanced models with great efficiency.

PDE-Regularized Regression for Heterogeneous Air Quality Data

Marco Francesco De Sanctis
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This work introduces a novel semiparametric approach for modeling spatio-temporal data characterized by grouping structures and complex physical dynamics. The method extends classical mixed-effects models by incorporating a nonparametric component regularized through a Partial Differential Equation. This formulation embeds prior physical knowledge, such as transport-diffusion phenomena, while random effects capture group-specific variability. Estimation relies on a functional Iterative Reweighted Least Squares algorithm coupled with an Expectation-Maximization step. Application to Nitrogen Dioxide concentrations in Lombardy demonstrates the model’s capacity to disentangle sensor-related noise from the underlying physical signal driven by wind fields.

Algebraic Issues Arising in Training Deep Operator Networks

Marco Dell’Orto

Università di Padova (Italy)

Empirical evidence shows that some operator networks trained to solve PDEs compress most of the learned information into a small number of dominant singular components (spikes) in their weight matrices. Random Matrix Theory provides tools to interpret the spectral structure of these weights. Further experiments show that the rank of the dataset also affects the spectral structure of the weight matrices. In this poster we present some results on how the rank of the teacher model and of the training dataset influence the rank of the trained student model.

Modeling Anisotropy via PDE Penalization: Application to Urban Mobility

Ilenia Di Battista

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This work proposes a semiparametric spatio-temporal modeling framework to address non-stationarity and anisotropy through differential penalization. The method introduces spatially varying anisotropy by penalizing a second-order linear partial differential equation, allowing local adaptation to complex spatial structures such as road networks. Non-stationary diffusion is modeled via diffusion tensors, handled using the Log-Euclidean metric for efficient tensor operations. The approach is illustrated using the Telecom Italia dataset on mobile phone activity in Milan, which exhibits strong directional patterns driven by the city’s road infrastructure, demonstrating the effectiveness of the proposed framework in capturing realistic spatio-temporal dynamics.

Reconstruction of Cardiac Lead Fields Using Auto-Decoders

Arsenii Dokuchaev

Università di Trento (Italy)

Solving inverse problems in cardiac electrophysiology requires repeated evaluation of forward models that map cardiac electrical activity to body-surface potentials. In the lead-field formulation, this mapping depends on patient-specific anatomical information, including torso conductivities, heart geometry and electrode placements. Although numerically accurate, lead-field computation is computationally demanding and often impractical in clinical practice, where detailed anatomy may be unavailable. Consequently, simplified pseudo-lead approximations are still commonly used, at the cost of reduced fidelity. We introduce a surrogate modeling approach for lead-field reconstruction based on an auto-decoder neural architecture. Our method learns a compact latent representation of lead fields conditioned on anatomical positional encoding and electrode locations, enabling rapid prediction for arbitrary electrode configurations, including standard ECG leads. Compared to classical numerical solvers, the surrogate dramatically reduces computation time and memory requirements, while delivering substantially improved results over pseudo-lead approximations. When used for ECG signal prediction, the surrogate produces excellent agreement with reference simulations. These findings demonstrate that learned lead-field surrogates are a computationally efficient, clinically relevant tool for patient-tailored

forward models, even with limited anatomical information. Our approach advances accessible digital twin workflows.

Time-continuous latent dynamics learning for nonlinear reduced order modeling

Nicola Farenga

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At the core of recently proposed deep learning (DL) –based approaches for the construction of data-driven reduced order models (ROMs), within a time-dependent multi-query context, lie two main components: autoencoders addressing the task of nonlinear dimensionality reduction, and an additional mechanism to model the temporal evolution of the resulting reduced representation – namely the latent dynamics – using regressive, autoregressive or recurrent approaches. Despite their effectiveness, DL-based ROMs are intrinsically tied to the temporal discretization employed during the offline training phase, consequently limiting their ability to adapt to different temporal resolutions during the online phase. This represents a critical constraint due to the need to undergo a fine-tuning or re-training stage whenever a new temporal discretization is encountered, resulting in additional offline computational costs. To this end, a central aspect in data-driven reduced order modeling resides in whether the learned ROM solution is a meaningful time-continuous approximation of the full order model (FOM) solution, so that the ROM can handle finer temporal discretizations than those encountered during the training stage by retaining a prescribed level of accuracy. We address this problem by introducing the latent dynamics model (LDM) mathematical framework, formalizing the task of latent dynamics learning by reframing the construction of a ROM from a dynamical perspective, in order to preserve the FOM’s initial value problem structure. The introduced formulation allows us to identify the different sources of approximation error associated to the time-discretization of the problem and the consequent adoption of learnable maps. Specifically, error and stability estimates for the LDM approximation of the FOM solution are derived, and the role of explicit Runge-Kutta schemes for the numerical integration of the latent dynamics is studied. In a learnable setting, where deep neural networks are employed to parameterize the LDM components, thus leading to a latent Neural ODE setting, an approximation result demonstrating LDMs’ ability to approximate the time-continuous dynamics is provided.

Toward Scalable and Valid Conditional Independence Testing with Spectral Features

Alek Fröhlich

Italian Institute of Technology (Italy)

Existing CI tests often rely on restrictive structural conditions, limiting their validity. Kernel methods using the partial covariance operator offer a more principled approach but suffer from limited adaptivity, slow convergence, and poor scalability. In this work, we explore whether representation learning can help address these limitations. Specifically, we focus on representations derived from the singular value decomposition of the partial covariance operator and use them to construct a simple test statistic. We also introduce a practical bi-level algorithm to learn these representations. Our theory links representation quality to test performance and establishes asymptotic validity and power guarantees.

Patient Specific Prediction of Glioblastoma Growth via Reduced Order Modeling and Neural Networks

Sara Gazzoni

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Glioblastoma (GBL) is an aggressive brain tumor whose invasion patterns are strongly shaped by patient-specific brain microstructure. We propose a proof-of-concept framework for patient-specific modeling of GBL growth, combining a diffuse-interface mathematical model with reduced-order modeling and neural networks. A proper orthogonal decomposition strategy is trained on synthetic data generated from patient-specific brain anatomies reconstructed from MRI and DTI. A neural network surrogate learns the inverse mapping from tumor evolution to model parameters, enabling real-time parameter estimation with high accuracy and significant computational speed-up. These results establish a methodological foundation for future clinical translation of patient-specific digital twins in neuro-oncology.

Random Set Point Processes for Space Mining

Alfredo Gimenez Zapiola

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Before launching a mission to mine a celestial body, it is imperative to possess a reliable mineralogical map of it. We propose to solve this by applying a new model using Spatial Point Processes and Random Set Theory. We work with the Cuprite (NASA) image. On a given pixel there is a point if the correlation of the observed spectrum with respect to known spectra of candidate minerals is above a high threshold. To help space miners, we model both the centers of mineral deposits and their shape: we achieve this through an MCMC-estimated point process with (random) set marks.

A Multiscale Kinetic–Hydrodynamic Framework for Image Segmentation

Giulia Guicciardi

Università degli studi di Pavia (Italy)

We extend a kinetic framework for consensus-based image segmentation, modeling pixels as particles with spatial positions and feature vectors evolving under compromise interactions and external noise. By coupling the spatial motion of particles with the evolution of their feature vectors, we derive the corresponding kinetic formulation. Using kinetic theory, macroscopic quantities are obtained by aggregating pixel-feature distributions. This enables a first-order hydrodynamic model via appropriate closures, embedding microscopic interaction parameters into a computationally efficient macroscopic formulation suitable for parameter optimization. Numerical examples illustrate the performance of this hierarchical methodology across the various scales.

A Feasibility-Correction Framework for Learning-Based Optimization

Fadil-Rahman Intahim

Georgia State University (Georgia)

We propose a feasibility-correction framework for learning-based optimization, a novel method that ensures neural predictions satisfy hard constraints. Our approach integrates a differentiable correction step that iteratively minimizes constraints violations starting from the network’s output, guaranteeing feasibility solutions. The entire pipeline is trained end-to-end, encouraging the network to make predictions that are near-feasible and near-optimal. We provide theoretical guarantees on feasibility and convergence. Empirically, the framework gives feasibility solutions with optimality gaps below 0.2% on convex problems and often finds better local optima in convex settings, while achieving 3-1000x speedups over the traditional solvers.

Mathematical Modeling of Cardiogenic Shock: Towards a Patient-Specific Simulation Framework for Therapy Optimization

Bianca Maria Laudenzi

Università di Trento (Italy)

Cardiogenic shock (CS) is a severe condition marked by ventricular failure, leading to low cardiac output and inadequate tissue oxygenation, with high mortality (50–80%) and heterogeneous treatment guidelines. To investigate the complex pathophysiology of CS and support future therapeutic decision-making, we present the development of a mathematical modeling framework aimed at simulating cardiovascular bio-signals under CS and predicting responses to therapeutic interventions. Clinically relevant bio-signals are simulated using a 0D global closed-loop cardio-respiratory model, whose parameters are calibrated through sensitivity analysis to reproduce both baseline and representative CS scenarios. Future developments will enable in silico exploration of treatment strategies, toward patient-specific clinical decision support.

Symmetric Convolutional AutoEncoders for model order reduction

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Autoencoders (AEs) have emerged as powerful tools for non-linear dimensionality reduction, often surpassing traditional linear methods such as Proper Orthogonal Decomposition (POD) in scenarios characterized by slowly decaying Kolmogorov n -widths. In the realm of Reduced-Order Modelling (ROM), these models are increasingly utilized to learn low-dimensional representations of solution manifolds associated with parametric Partial Differential Equations (PDEs). However, the high expressivity of AEs presents a challenge: although trained networks typically minimize reconstruction error, they often struggle to capture the essential properties necessary for building accurate ROM. To overcome this issue, it is crucial to introduce bias during training, ensuring the model learns a pair of encoder and decoder that maintains representation consistency—specifically, constraining the composition of the encoder and decoder to be equal to the identity map $E \circ D = id$ within the local coordinates of the chart formed by the encoder and the solution manifold. Recent works have tackled this challenge in fully connected AEs by proposing representation-consistent architectures. This study

builds upon that concept by extending representation consistency into the convolutional domain. We introduce a novel class of symmetric Convolutional Autoencoders (CAEs) designed to embody the primary properties of manifold parametrization mappings. When integrated into a ROM framework, this architecture demonstrates significantly improved predictive capabilities. Numerical experiments indicate that, in comparison to standard CAEs, our proposed approach achieves reduced reconstruction errors and generates more accurate latent-space trajectories.

Modelling of the interplay between vascular architecture and cancer cell plasticity under radiotherapy

Cristina Macaluso

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This study presents a computational model to investigate how hypoxia influences cancer cell plasticity and radiotherapy resistance. Calibrated using vascular data from a cohort of cancer patients, the framework couples cell distribution dynamics with oxygen transport models. Numerical simulations compare normoxic and hypoxic scenarios, revealing that while normoxia promotes differentiation, hypoxia allows radioresistant stem cells to thrive in specific niches. This approach successfully integrates patient data with multiscale physics to predict phenotypic evolution and optimize treatment efficacy against tumor aggressiveness.

A multi-agent digital twin for decision-making in market dynamics through active inference

Francesco Maria Mancinelli

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Active Inference (AIF) is a neuroscience-inspired framework for decision-making under uncertainty, integrating perception, learning, and planning within a unified Bayesian paradigm. Unlike reinforcement learning, AIF does not require predefined reward functions and naturally balances epistemic (information-seeking) and pragmatic (goal-directed) behaviours, making it well-suited for adaptive, real-world applications. This work explores the application of AIF to Digital Twin scenarios, where virtual models of socio-economic agents are continuously updated through real-time data to support monitoring, prediction, and decision-making. We extend AIF to multi-agent systems, enabling decentralized coordination through shared probabilistic beliefs, inspired by self-organizing biological systems. This approach addresses key challenges in multi-agent settings, including credit assignment, scalability, and the integration of epistemic actions. Streaming machine learning techniques are incorporated to dynamically adapt goal priors, allowing agents to anticipate and respond to evolving environments without fixed temporal assumptions. As a testbed, we introduce an extended Cournot model in which each firm is represented by an AIF agent responsible for production and warehouse management over an extended horizon. This setup enables the study of competitive and cooperative dynamics, emergent collective behaviour, and strategic adaptation within a principled game-theoretic framework. In particular, the model provides insights into social phenomena such as the bandwagon effect, illustrating how agents balance conformity with internally motivated epistemic and goal-directed strategies. Numerical simulations demonstrate that multi-agent AIF enables robust, predictive, and adaptive decision-making in partially observable market environments, outperforming traditional equilibrium-based approaches in terms of flexibility and information efficiency. By situating the analysis at the intersection of Digital

Twin research, game theory, and computational social science, this study presents a novel methodology for modeling and understanding complex socio-economic dynamics through active inference-driven virtual agents.

Thermodynamics Reveals the Generalization in the Interpolation Regime

Erfan Mirzaei

Italian Institute of Technology (Italy)

Understanding why overparameterized models generalize remains a central question in modern machine learning, especially in regimes where models interpolate the training data—even when labels are random. In this talk, I will present recent work that provides data-dependent generalization bounds for the Gibbs algorithm in the overparameterized interpolation regime. Our results show that meaningful control of the test error is still possible, despite the ability of the model to fit arbitrary data. A key feature of the proposed bounds is their stability under approximation by Langevin Monte Carlo algorithms, making them relevant for practical training procedures. Through experiments on MNIST and CIFAR-10, we demonstrate that the bounds yield nontrivial predictions for real labeled data, while correctly upper-bounding the test error when labels are randomized. Perhaps surprisingly, our analysis suggests that generalization at low temperature in the interpolation regime is already foreshadowed by small training errors in the classical high-temperature regime, offering a new perspective on the role of temperature and implicit regularization in modern learning dynamics.

Infinite-Dimensional Generative Compressed Sensing via Coherence-Based Sampling

Andrea Munno, Jacopo Senoner

Politecnico di Milano (Italy)

We present a comparative study of primal and mixed finite element discretizations of the power-law model for non-newtonian incompressible flows, covering both shear-thinning and shear-thickening regimes. We assess the performance of the MINI, lowest order Hood-Taylor (P2–P1), and the Kouhia-Stenberg elements. The resulting nonlinear problem is solved using different linearization strategies, depending on the exponent of the power-law model: fixed point iterations (Picard/Kačanov) and the Newton method. In the shear-thinning regime, where the effective viscosity may become strongly degenerate, we employ and investigate three regularization/relaxation techniques.

Hybrid resolution strategies for reduced models with high Kolmogorov n-width

Davide Oberto

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We will present some preliminary results on solving in a hybrid manner the linear system at the reduced level of models obtained from Galerkin-projection of the full order model. In particular, we will focus

on problems showing high Kolmogorov n -width requiring a high number of reduced basis to guarantee accuracy. In this setting, the resulting reduced linear system is high-dimensional and dense, making its resolution with classic direct methods costly. Our method proposes to solve only few dimensions with direct methods and to use iterative ones for the remaining dimensions, reducing the computational cost.

Generative models for Shape Deformation and Reduced Order Modelling

Guglielmo Padula

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Highly accurate simulations of geometry-parameterized systems require adaptable but limited shape representations. Free Form Deformation (FFD) and reduced-order models encounter difficulties due to high-dimensional parameter spaces, and the enforcement of constraints. This study presents constrained generative models to learn distributions of 3D forms while minimizing the parameter dimensionality through latent spaces. In fixed topology meshes, autoencoding models utilizing PCA and precise constraint projection maintain invariants like barycentre and volume. For general topologies, generative models are obtained by integrating latent codes with point coordinates. Combined with PPINNs and non intrusive reduced order models, the framework delivers precise, real-time PDE solutions.

A coupled mathematical and numerical model for protein spreading and tissue atrophy applied to Alzheimer’s disease

Valentina Pederzoli

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This work introduces and validates a new mathematical framework for modeling biological tissue atrophy driven by the diffusion of a biological agent, with emphasis on neurodegenerative diseases. The model couples a Fisher–Kolmogorov equation describing agent diffusion with an elasticity equation accounting for tissue mass loss, linked through a logistic law governing progressive atrophy. Applied to Alzheimer’s disease, it captures tau-protein propagation and resulting brain atrophy. A Discontinuous Galerkin method and Crank–Nicolson scheme are used for numerical solution, with convergence studies and simulations validating the model’s ability to reproduce key disease features.

Bayesian Ensemble Forest for Anomaly Detection

Federico Maria Quetti

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Random Forest was extended from classification/regression tasks to anomaly detection through the Isolation Forest algorithm. Such ensemble learning techniques rely on bootstrap aggregation, which provides accuracy and robustness but is limited in its ability to incorporate prior information, operating purely on empirical data distributions. We instead exploit a posterior-resampling bootstrap scheme,

thus developing a prior-sensitive adaptation of the Isolation Forest algorithm where trees are defined through posterior draws, bridging Bayesian nonparametrics and unsupervised learning. We explore the specification of suitable prior distributions, assessing the proposal's performance on real datasets, demonstrating its validity and potential advantages in anomaly detection tasks.

Nonparametric estimation of conditional probability distributions using a generative approach based on conditional push-forward neural networks

Lorenzo Tedesco

Università di Bergamo (Italy)

We introduce conditional push-forward neural networks (CPFNN), a new generative approach for estimating conditional distributions. Instead of trying to model the conditional probability of a target variable directly, CPFNN learns a stochastic transformation that takes in an input x along with a random latent vector and produces outputs that behave like the true outcomes associated with x . This design makes it easy to generate conditional samples and to estimate conditional statistics using simple Monte Carlo sampling. Training relies on an objective inspired by information-theoretic principles, but it does not require invertible models or adversarial training. We show that, as the amount of data grows, the method becomes nearly consistent with the true underlying distribution. Experiments demonstrate that CPFNN can match or outperform leading methods—including kernel models, tree-based approaches, and modern deep learning techniques—while remaining lightweight and easy to train.

Structure Preserving Multiscale Methods for Supercritical Fokker-Planck Equations

Horacio Tettamanti

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We focus on a specific type of Fokker-Planck equation including a nonlinear drift term used to describe a variety of systems ranging from opinion models to the dynamics of quantum particles. In all of these settings it has been proved that depending on the choice of the model parameters the equilibrium state of the system is characterized by a condensate state leading to the formation of a singularity described by the emergence of a Dirac Delta. It has been shown that in the presence of the condensate state the solution loses L^2 regularity and therefore its propagation has to be studied by introducing a weighted space where new Nash and Gagliardo-Nirenberg inequalities have been developed to show that the solution blows-up in finite time. We focus on the design of structure preserving numerical methods that are able to capture the different behaviors of the solution in the presence and absence of the singularity while preserving important properties such as entropy decay and positivity of the solution. We show that our method is able to correctly capture the emergence of the condensation effects both in a one-dimensional and two-dimensional setting. To this end, we introduce a novel hybrid scheme where we couple our numerical scheme with a particle system enabling us to deal with the formation of the Dirac Delta.

A stochastic investigation of bifurcating phenomena via Polynomial Chaos Expansion

Giacomo Venier

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Bifurcating phenomena in parameter-dependent differential equations can cause qualitative changes in the system dynamics, often leading to ill-conditioning. This work employs generalized Polynomial Chaos Expansion (PCE) to study these systems from an Uncertainty Quantification perspective, focusing on one-dimensional dynamical system normal forms. Two types of solutions emerge: oscillating and branch-approximating, both of which enable the development of tools for recovering bifurcation diagrams, with theoretical justification and numerical validation. This framework provides insights into the statistical and analytical properties of PCE solutions and their connection to underlying bifurcation structures, with potential extensions to higher-dimensional and more complex systems, such as PDEs.

Evaluating the Accuracy of Monocular 3D Pose Estimation for Gait Analysis

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Human gait analysis is essential in clinical and rehabilitation settings for assessing neuromotor function and functional impairments. Although artificial intelligence has significantly advanced markerless motion capture, the use of monocular video-based approaches for biomechanically informed gait assessment remains limited. This study investigates the feasibility of monocular 3D human pose estimation for extracting clinically relevant gait features. We evaluate two monocular models, RTMPose3D and BlazePose3D, by computing sagittal-plane hip, knee, and ankle joint angles through an OpenSim-based biomechanical workflow. Their outputs are compared against a reference multi-view 3D pose estimation pipeline. Results show that while monocular methods successfully capture overall gait dynamics, notable discrepancies in joint kinematics persist when compared to multi-view estimates. These preliminary findings highlight both the promise and current limitations of monocular pose estimation for gait analysis, indicating the need for broader validation, particularly in clinical populations, before adoption in real-world rehabilitation contexts.

Multi-Fidelity SINDy: Sparse Discovery of Nonlinear Dynamical Systems with Fidelity-Weighted Measurements

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Data from simulations and experiments are rarely noise-free and often exhibit varying levels of fidelity. Measurement uncertainty may differ across repeated observations, sensor apparatus, or even during a single experiment. This work addresses the problem of discovering nonlinear dynamical systems from such inhomogeneous data. We extend the Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework to handle variable noise levels by combining Ensemble SINDy and Weak SINDy within a weighted regression formulation derived from weighted least squares. The approach not only

mitigates the adverse effects of heteroscedastic noise but also shows that repeated, cost-effective, low-quality measurements can enhance model recovery, achieving performance comparable or superior to reconstructions based solely on high-fidelity data. A mathematical intuition for the method’s optimality is provided. The study is organized in two parts. The first examines scenarios with multiple trajectories of the same dynamical system, each affected by distinct noise levels, highlighting the benefits of their integration. The second focuses on single-trajectory data with spatially or temporally varying noise. The methodology is validated on several benchmark systems, including the Lorenz attractor, the double pendulum, the Hopf oscillator, Burgers’ equation, and the Navier–Stokes equations. The results demonstrate that properly weighted sparse regression transforms measurement heterogeneity from a limitation into a source of information, enabling robust and accurate model discovery in complex dynamical systems.



Program of the event